

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705I - BRIDGE ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. Bearings in a bridge are used to
 - a. Support the deck directly
 - b. Increase span length
 - c. Provide aesthetic appeal
 - d. Transfer loads from superstructure to substructure
 2. A bridge classified by location as “causeway” is built:
 - a. Across a deep navigable channel requiring high-level clearance
 - b. Along a steep embankment to connect two elevated hill slopes
 - c. Across shallow rivers with low embankments
 - d. Over a dry terrain where no water flow is expected
 3. Waterway calculation ensures:
 - a. Traffic volume management
 - b. Soil stability
 - c. Adequate width for flood flow passage
 - d. Proper deck thickness
 4. Freeboard in a river bridge is provided to:
 - a. Increase soil strength
 - b. Allow clearance above high flood level
 - c. Reduce scour depth
 - d. Increase deck thickness
 5. Longitudinal forces in road bridges are mainly due to:
 - a. Braking and tractive forces of vehicles
 - b. Wind loads
 - c. Dead load
 - d. Buoyancy

- 6 Impact factor accounts for:
 - a. Seismic forces
 - b. Dynamic effect of moving loads
 - c. Dead load
 - d. Buoyancy
- 7 Grillage analysis models the bridge deck as:
 - a. Hollow box section
 - b. Grid of beams
 - c. Prestressed slab
 - d. RCC wall
- 8 Box girder bridges are most suitable for:
 - a. Short spans only
 - b. Long spans with high torsional rigidity
 - c. Pedestrian bridges only
 - d. Temporary bridges
- 9 Parapets in highway bridges are mainly provided for:
 - a. Aesthetic appeal
 - b. Safety of vehicles and pedestrians
 - c. Reducing dead load
 - d. Increasing freeboard
- 10 Well foundations are commonly used for:
 - a. Small culverts
 - b. Temporary structures
 - c. Deep river crossings and heavy loads
 - d. Footpaths
- 11 Wing walls are provided to:
 - a. Increase foundation depth
 - b. Retain earth at bridge approaches
 - c. Reduce dead load
 - d. Increase pier height
- 12 The part of an abutment that resists horizontal earth pressure is:
 - a. Cap
 - b. Backwall
 - c. Toe slab
 - d. Curtain wall

Q2. Solve the following.

- A) Explain with neat sketch the components of a bridge. 6
- B) Explain the classification of road bridges based on span length. 6

Q3. Solve the following.

- A) Explain the factors affecting the selection of a road bridge site. 6
- B) State the factors governing the depth of bridge foundation. 6

Q4. Solve Any Two of the following.

- A) Define dead load and live load as per IRC specifications. How are they estimated for road bridges? 6
- B) What are the typical traffic loading patterns used for road bridges? 6
- C) Discuss the effect of centrifugal force and wind load on bridge structures. How are these loads considered in design? 6

- Q5. Solve Any Two of the following.
- A) What are expansion joints in bridges? Describe their necessity and various types used in modern bridge construction. 6
 - B) Define bridge parapet. Explain the classification and functional requirements of parapets for highway bridges. 6
 - C) Write short notes on:(a) Orthotropic plate theory (b) Grillage analogy 6
- Q6. Solve Any Two of the following.
- A) What are abutments? Describe their functions and the loads they resist. 6
 - B) State the importance of soil–structure interaction in bridge design. How does it affect foundation performance? 6
 - C) Explain the purpose of wing walls. How are they classified and designed? 6

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